

Bypass Root Detection – Complete Guide

Root detection bypass ka matlab hai app ko yeh believe karwana ke device rooted nahi hai, jabke actually hai. Yeh dynamic analysis aur penetration testing mein ek common requirement hai kyunke most tooling root access maangti hai . Banking apps, financial apps, aur security-sensitive apps aksar rooted devices ko block kar deti hain .

Root Detection Kaise Kaam Karti Hai?

Apps root detect karne ke liye multiple techniques use karti hain :

File-based Checks: Common root-related files aur directories check karti hain jaise `/system/xbin/su`, `/system/bin/su`, `/data/adb/magisk`, `/sbin/.magisk`

Package-based Checks: Root management apps (Magisk, SuperSU, KingRoot) ke package names scan karti hain

Shell Command Checks: `su`, `which su`, `mount`, `getprop ro.debuggable` jaise commands execute karke root detect karti hain

System Property Checks: `ro.debuggable = 1`, `ro.secure = 0`, `ro.build.tags = test-keys` jaise properties check karti hain

Method 1: Objection (Quickest & Easiest)

Objection ek Frida-based runtime exploration toolkit hai. Iska command root detection bypass karne ke liye built-in hai .

Command:

```
objection -g com.example.app explore  
android root disable
```

Yeh command common root detection APIs ko hook karke false ya safe values return karta hai . Objection bina jailbreak/root ke bhi kaam kar sakta hai .

Method 2: Frida Universal Scripts

Option A: Comprehensive Bypass Script

(Recommended)

```
frida -U -f com.example.app -l frida-android-  
rootandpinning.js --no-pause
```

Yeh script combine karti hai:

- Package checks ko bypass karna (Magisk, SuperSU packages hide)
- Binary checks (su, busybox non-existent dikhana)

- Property overrides (ro.secure, ro.debuggable spoof karna)
- Command execution intercept karna (id, mount, getprop)

Option B: GitHub se Script Download Karein

Multiple repositories available hain jinmein ready-to-use scripts hain:

secfb/Frida-Android-SSL-Pinning-Script – Multiple scripts hain jo root + SSL bypass ek saath karti hain

g2salam-91/Frida-script – Comprehensive anti-detection bypass tool hai jo root, emulator artifacts, aur SSL pinning ko hide karta hai

riyadmondol2006/Android-Root-Detection-Bypass – Native aur Java dono level par hooks provide karta hai

Method 3: Magisk-Based Setup (For Live Apps)

Banking apps jaise sensitive apps ke liye, permanent solution Magisk modules ke zariye hota hai :

Step 1: Initial Setup

- Magisk app ko hide karein (Settings → Hide the Magisk app)
- Saare conflicting modules disable karein

Step 2: Required Modules Install Karein

- Play Integrity Fork – MEETS_BASIC_INTEGRITY pass karne ke liye
- Shamiko – Zygisk-based root hiding
- Zygisk Assistant – Open-source alternative
- HideMyApplist – Root apps ko target se chhupane ke liye

Step 3: Configure Magisk Settings

- Zygisk toggle OFF karein (Shamiko handle karega)
- Enforce DenyList OFF karein
- Configure DenyList mein target app aur uske sub-processes add karein

Step 4: HideMyApplist Configure

- Blacklist template banayein
- Root apps (Magisk, LSPosed, file explorer) add karein
- Target app par apply karein

Method 4: LSPosed Module

Xposed-based approach hai jo Java aur Native C/C++ dono level par hooks provide karta hai .

Bypass Root Check Pro – RootBeerFresh detections, SELinux test, R/W System check, su/magisk/busybox binaries, root-related apps, build test keys, props, root cloaking apps, aur Magisk Unix Domain Socket sab bypass karta hai .

Method 5: Manual Frida Script (Custom Implementation)

Agar aapko specific method bypass karna hai toh manually hook kar sakte hain :

```
Java.perform(function () {  
    var targetClass =  
Java.use("com.example.RootChecker");  
    targetClass.isDeviceRooted.implementation =  
function () {  
        console.log("[*] isDeviceRooted() called -  
returning false");  
        return false;  
    };  
});
```

JADX se code analyze karo, root detection classes identify karo, aur unke methods ko false return karne ke liye override karo .

Play Integrity API (For Banking Apps)

Bohot banking apps Play Integrity API use karti hain MEETS_BASIC_INTEGRITY check karne ke liye. Agar fail ho toh root detect ho jata hai .

Solution: Play Integrity Fork module install karein aur "Delete Storage" + "Check" process repeat karein jab tak MEETS_BASIC_INTEGRITY pass na ho .

Important Notes

Root Required: Most Frida-based methods require rooted device aur Frida server root permissions ke saath chalna chahiye .

Version Matching: PC aur phone par Frida versions match honi chahiye .

No Guarantee: Root detection bypass "cat and mouse game" hai – success never guaranteed aur methods time ke saath obsolete ho sakti hain .

Testing: RootBeer Fresh sample app use karke apne setup ko test kar sakte hain .

 **Educational Purpose Only:** Root detection bypass sirf apne own devices, authorized penetration testing, aur bug bounty programs mein karein. Kisi doosre ke app mein bina permission ke bypass karna applicable cyber laws ke under punishable hai.

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